

# MICHAEL CHO

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## RESEARCH INTERESTS

The application of discrete math, probability, and complexity theory to various problems in theoretical computer science at the intersection of algorithms, combinatorics, and optimization.

Common interests include hardness of approximation and approximation algorithms for problems in matching theory, social welfare, and other topics in algorithmic game theory.

Further interest in related areas in computer science such as AI/ML and systems, as well as counting combinatorial objects — for example meanders, self-avoiding walks, and heapable permutations.

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## EDUCATION

### Stanford University

2021-2026

*Expected: Mathematics (BS), Computer Science (MS)*

Overall GPA: 3.88

Graduate courses: randomized algorithms, combinatorial optimization, complexity theory, theory of probability, algorithmic game theory, algebraic error-correcting codes, deep generative models, counting and sampling, operating systems, cryptography, natural language processing.

Undergraduate courses: discrete probabilistic methods, linear algebra, number theory, topology and geometry, galois theory, representation theory, graph theory, combinatorics, complex analysis, functional analysis.

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## RESEARCH EXPERIENCE

### Stochastic Matching with Batch Allocations

2024 - current

Defined a variation on stochastic matching problems with agents receiving “batches” of items. Proved hardness of approximation results using novel constructions and gave approximation algorithms beating the classic  $1 - 1/e$  approximation ratio. Honors thesis and research guided by Jan Vondrak at Stanford.

### Approximation Algorithms for Online Stochastic Matching

2023 - 2024

Studied existing and novel algorithms for the online bipartite matching problem when constrained to polynomial time. Developed computational bounds and rigorous proofs for performance against the optimal online benchmark. Research guided by Professor Amin Saberi and Tristan Pollner at Stanford.

### Optimal Constructions for DNA Self-Assembly of $k$ -Regular Graphs

2023

Created novel bounds and constructions within a graph theoretical model of nanostructure self-assembly. Used computational tools, combinatorics, and algebra to analyze optimal configurations and prove useful properties. Group project guided by Summer@ICERM 2023 organizers, accepted for publication in *Involve*.

### Counting the Number of Meanders

2023

Studied the best known asymptotic bounds and methods of computing the number of meanders—a combinatorial object consisting of a self-avoiding closed loop crossing a straight line. Examined and disproved a related conjecture using computational methods. Directed reading guided by Jacopo Borga at Stanford.

### Efficient Methods of Calculating the Number of Heapable Permutations

2020 - 2021

Discovered and proved novel results in a variation on the Longest Increasing Subsequence problem based on heaps, a graph data structure. Improved time complexity of best-known algorithm for computing the number of heapable permutations. Group project at PROMYS published in *Discrete Applied Mathematics*.

### Given Certain Conditions, Even Selfish Prisoners Can Learn to Cooperate

2019 - 2020

Studied optimal strategies for the Iterated Prisoner’s Dilemma through computer simulations. Used the  $n$ -player version of the game and computational game theory to theorize conditions in which cooperation

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<sup>1</sup>Updated November 17, 2025

naturally arises. Independent project published in ActonScholars, an online high school journal.

### **Game of Life to Simulate Computers**

*2019 - 2020*

Investigated methods of constructing logic gates and memory within Conway's Game of Life, a cellular automaton known to be Turing Complete. Used glider mechanics to simulate basic logic. Independent project published in iJournals, an online high school journal.

## **WORK EXPERIENCE**

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### **Software Engineering Intern — Mintlify**

*2025*

Worked on the Product Engineering team to build customer-facing functionality and address user concerns. Designed and implemented a new page navigation framework to improve DevX and support modern features.

### **Quantitative Trading Intern — Citadel Securities**

*2024*

Worked on the Block Options and Semi-Systematic Options desks. Studied overnight volatility surface and term structure movements of both index and single-stock options to supplement trader intuition.

## **TEACHING EXPERIENCE**

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### **Course Assistant for CS166: Advanced Data Structures at Stanford University**

*2025*

Assisted in grading assignments, writing exams, and answering student questions through office hours and on-line forums for a 70+ student course. Held weekly sessions to supplement lectures, which covered a variety of topics (amortization, randomization, RMQ, string data structures, etc). Received 4.6/5 rating from students.

### **Math and Coding Club Tutor at Seoul International School**

*2019 - 2021*

Used leadership positions in the coding and math clubs to tutor 30+ students in various subjects including calculus, discrete mathematics, and algorithms. Organized weekly workshops for talks and collaborative work on algorithmic problems.

### **Volunteer English Tutor at Seongnam Migrant Center**

*2017 - 2021*

Prepared lesson plans and taught 2-hour weekly lessons for elementary school children without access to upper-level English education in Korea. Worked with organizers to tailor lessons for children of various levels of English experience with focus on advanced schoolwork and practical use.

## **EXPERIENCE IN TALKS AND PRESENTATIONS**

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### **Efficient Matching Under Distributional Constraints**

*October 2025*

Stanford University

### **Expander Error Correcting Codes**

*March 2025*

Stanford University

### **Textual Style Transfer for Poetry**

*December 2024*

Stanford University

### **Optimal Constructions for DNA Self-Assembly of $k$ -Regular Graphs**

*January 2024*

Joint Mathematics Meetings

### **Counting the Number of Meanders**

*December 2023*

Stanford University

### **Optimal Constructions for DNA Self-Assembly of $k$ -Regular Graphs**

*August 2023*

Brown University Summer Research Symposium

### **Optimal Constructions for DNA Self-Assembly of $k$ -Regular Graphs**

*August 2023*

Summer at the Institute for Computational and Experimental Research in Mathematics

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| <b>Counting the Number of Heapable Permutations</b><br>Stanford Undergraduate Mathematics Organization Colloquium | <i>June 2023</i> |
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| <b>Counting and <math>\#P</math>, a Story Parallel to <math>P</math></b><br>Stanford University | <i>June 2023</i> |
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| <b>Self-Avoiding Walks and the Hammersley-Welsh Method</b><br>Stanford University | <i>March 2023</i> |
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| <b>Bounding and Computing the Number of Heapable Permutations</b><br>Program in Mathematics for Young Scientists | <i>August 2020</i> |
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| <b>Approximating Solutions to Diophantine Systems of Equations</b><br>Program in Mathematics for Young Scientists | <i>August 2019</i> |
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#### WORKSHOPS AND SEMINARS

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| <b>BCV Labs Summer Fellowship</b> | <i>2025</i> |
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| <b>North America Programming Camp</b> | <i>2025</i> |
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| <b>Joint Mathematics Meetings</b> | <i>2024</i> |
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| <b>Summer@ICERM Institute for Mathematical Modeling of DNA Self-Assembly</b> | <i>2023</i> |
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| <b>ECOPRO Summer School, Institute of Basic Science</b> | <i>2022</i> |
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| <b>Stanford Competitive Programming Club</b> | <i>2021 -</i> |
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| <b>Program in Mathematics for Young Scientists (PROMYS)</b> | <i>2019, 2020</i> |
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| <b>Worldwide Online Olympiad Training</b> | <i>2018 - 2019</i> |
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#### AWARDS IN MATHEMATICS AND ALGORITHMS

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| <b>Rank 5 out of 69 at ICPC PACNW Regional 2024,<br/>represented Stanford at ICPC North America Championships 2025</b> | <i>2024-2025</i> |
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| <b>Top 200 scorer, Putnam Competition 2023</b> | <i>2023</i> |
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| <b>Rank 681 out of 32702, Google Codejam 2022</b> | <i>2022</i> |
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| <b>Rank 375 out of 9004, Google Hash Code 2021</b> | <i>2021</i> |
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| <b>Rank 1236 out of 34585, Facebook Hacker Cup 2021</b> | <i>2021</i> |
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#### AWARDS IN WRITING

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| <b>Russell A. Berman Award for Excellence in an Introductory Seminar</b><br><i>Title: Exploring the Merits of Prior Acquaintanceship and Collaboration in Stanford Courses</i> | <i>2022</i> |
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| <b>Boothe Prize Nomination for Excellence in Writing</b><br><i>Title: Examining Inequalities and Ineffectiveness in Korea's English Education</i> | <i>2022</i> |
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## PUBLICATIONS

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- [1] Lisa Baek, Ethan Bove, Michael Cho, Xingyi Zhang, Leyda Almodóvar, Amanda Harsy, Cory Johnson, and Jessica Sorrells. *Optimal Constructions for DNA Self-Assembly of  $k$ -Regular Graphs*. 2025. arXiv: [2502.03716](https://arxiv.org/abs/2502.03716) [math.CO]. URL: <https://arxiv.org/abs/2502.03716>.
- [2] B. Chen, M. Cho, M. Tutuncu-Macias, and T. Tzolov. “Efficient methods of calculating the number of heapable permutations”. *Discrete Applied Mathematics* 331 (2023), pp. 126–137. URL: <https://authors.elsevier.com/c/1gX3j,3nuHslx8>.